

ANX-PR/CL/001-01

LEARNING GUIDE

SUBJECT

203000026 - Genomics Assited Breeding

DEGREE PROGRAMME

20BC - Master Universitario En Biología Computacional

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 1

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1. Description

1.1. Subject details

Name of the subject	203000026 - Genomics Assited Breeding
No of credits	3 ECTS
Type	Optional/elective
Academic year ot the programme	First year
Semester of tuition	Semester 1
Tuition period	September-January
Tuition languages	English
Degree programme	20BC - Master Universitario en Biología Computacional
Centre	20 - Etsi Agronómica, Aliment. Y Biosistemas
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Julio Isidro Sanchez (Subject coordinator)		j.isidro@upm.es	- -

* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

2.3. External faculty

Name and surname	Email	Institution
Julio Isidro Y Sánchez	j.isidro@upm.es	Centro de Biotecnología y Genómica de Plantas UPM-INIA

3. Prior knowledge recommended to take the subject

3.1. Recommended (passed) subjects

The subject - recommended (passed), are not defined.

3.2. Other recommended learning outcomes

- Although, the course is not aimed at students with advanced statistical skills, they should be familiar with statistical concepts. Course tutorials will use statistical packages in R, and experience in R is recommended but is not essential.

4. Skills and learning outcomes *

4.1. Skills to be learned

CE02 - Utilizar sistemas operativos, programas y herramientas de uso común en biología computacional, así como, manejar plataformas de cómputo de altas prestaciones, lenguajes de programación y análisis bioinformáticos

CE03 - Analizar e interpretar bioinformáticamente los datos que se derivan de las tecnologías ómicas, y proponer soluciones bioinformáticas en relación a dichos datos.

CE05 - Utilizar herramientas de biología computacional para el análisis genómico, incluida la genómica comparativa y biología evolutiva.

CE10 - Conocimiento de las técnicas de representación del conocimiento reutilizables y modelos de razonamiento en entornos centralizados y distribuidos a utilizar en la resolución de problemas que impliquen conducta inteligente.

CG02 - Familiarizarse con el trabajo y los métodos de la Biología Computacional en condiciones reales,

adquiriendo la capacidad de diseñar aplicaciones/experimentos de forma independiente y describir, cuantificar, analizar y evaluar críticamente los resultados obtenidos.

CG03 - Que los estudiantes sepan aplicar los conocimientos adquiridos y su capacidad de resolución de problemas en entornos nuevos o poco conocidos dentro de contextos más amplios (o multidisciplinares) relacionados con el área de la Biología Computacional.

CT03 - Tener compromiso bioético y profesional y respeto por la sostenibilidad ambiental.

4.2. Learning outcomes

RA13 - Conocer los últimos desarrollos, técnicas y herramientas en bioinformática y biología computacional aplicada a los programas de mejora genético

RA14 - Conocimientos para aplicar las nuevas tecnologías genómicas a proyectos reales de mejora genética de cultivos

RA15 - Conocimiento de las herramientas genómicas para caracterización de bancos de recursos genéticos y su utilización en programas de mejora.

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

5. Brief description of the subject and syllabus

5.1. Brief description of the subject

Advancements in the fields of genetics and genomics have provided profound insights into both the structural and functional dimensions of plant genomes. These revelations not only deepen our understanding but have also equipped us with novel tools and methodologies, facilitating more efficient and cost-effective research strategies in crop plant studies. To fully harness the potential of these innovations, there's an imperative to converge diverse disciplines like structural genomics, transcriptomics, proteomics, metabolomics, plant physiology, and plant breeding. A plant breeder's paramount objective is to analyze the genetic composition of individual plants, selecting those genotypes that exhibit the most promising traits within breeding populations. Early selection yields better genetic enhancement. The traditional marker-assisted breeding approach has now evolved into genomic assisted breeding, largely attributed to the plummeting costs of individual genotyping.

This course delves into Genomic Selection (GS) or Genomic Prediction (GP), a cutting-edge breeding methodology leveraging genome-wide molecular markers to predict breeding values for complex quantitative traits.

Our goal for this course is to furnish participants with a foundational understanding of the quantitative and statistical aspects essential for routine GS application. Emphasizing hands-on learning, this course will revolve around practical plant breeding exercises using the R platform.

Upon completing this course, participants can expect the following learning outcomes:

- Gain a foundational grasp of plant breeding and quantitative genetics in the context of enhancing crops.
- Acquire the essential knowledge to construct Genomic Selection/Genomic Prediction models from the ground up.
- Implement supervised machine learning techniques within plant breeding initiatives.

5.2. Syllabus

1. Introduction of Genomic Assisted Breeding.
2. Plant breeding overview.
3. Review of Quantitative genetics/ Quantitative trait loci.
4. Sources of quantitative trait variation.
5. Breeding Values and Heritabilities.
6. Resemblance among relatives.
7. Pedigree vs Kinship matrix.
8. Big Picture of Genomic selection/ Machine learning approach.
9. Factors affecting Genomic Selection.
10. Imputation.
11. Statistical Models
 - 11.1. Linear Models/ANOVA/GLM.
 - 11.2. Fixed-Random Effects.
 - 11.3. Best Linear Unbiased Estimator.
 - 11.4. Best Linear Unbiased Predictor.
 - 11.5. One step model.
 - 11.6. Two step models.

6. Schedule

6.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	Topic 1. Plant breeding overview. Review of Quantitative genetics/ Quantitative trait loci Duration: 02:00 Lecture			
2	Topic 1. Plant breeding overview. Review of Quantitative genetics/ Quantitative trait loci Duration: 02:00 Laboratory assignments			
3	Topic 1 Duration: 02:00 Additional activities			Topic 1 Individual work Progressive assessment and Global Examination Presential Duration: 00:00
4	Topic 2. Genomic selection/ Machine learning approach Duration: 02:00 Lecture			
5	Topic 2 Duration: 02:00 Laboratory assignments			
6	Topic 2 Duration: 02:00 Additional activities			Topic 2 Individual work Progressive assessment and Global Examination Presential Duration: 00:00
7	Topic 3. Plant breeding methods and genomics assisted breeding Duration: 02:00 Lecture			
8	Topic 3 Duration: 02:00 Laboratory assignments			
9	Topic 3 Duration: 02:00 Additional activities			Topic 3 Individual work Progressive assessment and Global Examination Presential Duration: 00:00

10	Topic 4. Statistical analyses for genomics assisted breeding Duration: 02:00 Lecture			
11	Topic 4 Duration: 02:00 Laboratory assignments			
12	Topic 4 Duration: 02:00 Additional activities			Topic 4 Individual work Progressive assessment and Global Examination Presential Duration: 00:00
13	Topic 5. Bioinformatics approaches in genomics assisted breeding Duration: 02:00 Lecture			
14	Topic 5 Duration: 02:00 Laboratory assignments			
15	Topic 5 Duration: 02:00 Additional activities			Topic 5 Individual work Progressive assessment and Global Examination Presential Duration: 00:00
16				
17				Global exam Written test Progressive assessment and Global Examination Presential Duration: 00:00

Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

7. Activities and assessment criteria

7.1. Assessment activities

7.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
3	Topic 1	Individual work	Face-to-face	00:00	8%	5 / 10	CG02 CG03 CT03 CE02 CE03 CE05 CE10
6	Topic 2	Individual work	Face-to-face	00:00	8%	5 / 10	CG02 CG03 CT03 CE02 CE03 CE05 CE10
9	Topic 3	Individual work	Face-to-face	00:00	8%	5 / 10	CG02 CG03 CT03 CE02 CE03 CE05 CE10
12	Topic 4	Individual work	Face-to-face	00:00	8%	5 / 10	CG02 CG03 CT03 CE02 CE03 CE05 CE10
15	Topic 5	Individual work	Face-to-face	00:00	8%	5 / 10	CG02 CG03 CT03 CE02 CE03 CE05 CE10

17	Global exam	Written test	Face-to-face	00:00	60%	5 / 10	CG02 CG03 CT03 CE02 CE03 CE05 CE10
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7.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
3	Topic 1	Individual work	Face-to-face	00:00	8%	5 / 10	CG02 CG03 CT03 CE02 CE03 CE05 CE10
6	Topic 2	Individual work	Face-to-face	00:00	8%	5 / 10	CG02 CG03 CT03 CE02 CE03 CE05 CE10
9	Topic 3	Individual work	Face-to-face	00:00	8%	5 / 10	CG02 CG03 CT03 CE02 CE03 CE05 CE10
12	Topic 4	Individual work	Face-to-face	00:00	8%	5 / 10	CG02 CG03 CT03 CE02 CE03 CE05 CE10
15	Topic 5	Individual work	Face-to-face	00:00	8%	5 / 10	CG02 CG03 CT03 CE02 CE03 CE05 CE10

17	Global exam	Written test	Face-to-face	00:00	60%	5 / 10	CG02 CG03 CT03 CE02 CE03 CE05 CE10
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7.1.3. Referred (re-sit) examination

Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
Examen escrito	Written test	Face-to-face	01:00	100%	5 / 10	CG02 CG03 CT03 CE02 CE03 CE05 CE10

7.2. Assessment criteria

Assessments will be based on your overall number of points earned out of the total possible points.

- Progressive evaluation: 40% of final grade.

- Global evaluation: 60% of final grade.

Further minor adjustments in final grades may be made at the discretion of the instructor based upon attendance and class participation. If you haven't passed the minimal grade to gain the learning outcomes of the module, a final exam will be provided. The final exam will last 120 minutes. Continuous assessment exercises will be submitted in Moodle unless otherwise is indicated. The instructions for the exam solutions will be given to the students when necessary.

The results will follow the scheme established by the UPM in 2024 as A: Excelent, B: Advanced, C: Satisfactory, D: Not satisfactory.

The primary venue for our classes will be on site at Informatics building in Montegancedo. However, please note that on certain occasions, due to unforeseen circumstances, sessions may transition to an online format. The main content for exam questions will be derived from the material covered during our classroom interactions. At times, questions may also source from textbook readings or other references highlighted during lectures. For theoretical portions of the course, we'll predominantly employ presentations and whiteboard tools to enhance understanding. On the practical side, labs will be centered around the R programming language. Throughout the duration of the course, class notes and pertinent materials will be shared with all participants via Teams.

This course emphasizes the practical application of plant breeding concepts, primarily through exercises in R. Consequently, the use of computers and proficiency in R programming is essential to grasp these concepts fully. To ensure a seamless learning experience, please install both R and RStudio on your computers prior to our sessions.

8. Teaching resources

8.1. Teaching resources for the subject

Name	Type	Notes
Principles of Plant Genetics and Breeding, 2 ed. by George Acquaah. 2012 Wiley-Blackwell Publishing, West Sussex, UK. ISBN 978-0-470-66475-9	Bibliography	
Falconer DS, Mackay TFC (1996) Introduction to quantitative genetics, 4th edn. Longman, New York	Bibliography	
Meuwissen THE, Hayes BJ, Goddard ME (2001) Prediction of total genetic value using genome-wide dense marker maps. Genetics 157:1819-1829	Bibliography	

Bernardo, R., 2002. Breeding for quantitative traits in plants (Vol. 1, p. 369). Woodbury, MN: Stemma press.	Bibliography	
Heslot N, Yang HP, Sorrells ME, Jannink JL (2012) Genomic selection in plant breeding: a comparison of models. Crop Sci 52:146-160	Bibliography	
Isidro, J., Jannink, J. L., Akdemir, D., Poland, J., Heslot, N., & Sorrells, M. E. (2015). Training set optimization under population structure in genomic selection. Theoretical and applied genetics, 128, 145-158.	Bibliography	
Isidro J, Akdemir D, Burke J (2017) Genomic selection. In: The world wheat book, pp 1001-1019	Bibliography	

9. Other information

9.1. Other information about the subject

La asignatura se relaciona con el ODS12.